

CSE 4411: Data Communication & Networking

Assignment (20 Marks)

Task

You will propose a realistic, implementable solution to a real-world problem that can be solved using networking concepts learned in this course. Your submission must include: problem description, requirements, protocol or technology choices with justification, and a practical plan for implementation and testing.

Report structure

1. **Title & Summary** (≤ 3 lines) – Short title and 2-3 sentence summary of the problem & proposed solution.
2. **Problem statement & Motivation** – Explain the real-world problem, affected stakeholders, and why a networked solution is needed. Clear context and rationale.
3. **Implementation & Testing Plan** – Describe a feasible proof-of-concept or deployment plan, list tools and test cases
4. **Feasibility, Cost & Impact Analysis** – Provide a brief cost estimate, deployment challenges, scalability issues, and social/environmental considerations.
5. **Presentation & Clarity** – Clear writing, labelled diagrams, proper referencing and formatting.

Submission format & guidelines

- **File format:** Single PDF (maximum **6 pages** A4, 11–12pt) plus up to **1 page** for diagrams if needed. If you include simulation code or configuration files, submit them in a ZIP alongside the PDF (optional).
- **Individual work:** Each student must submit an original proposal. Discussing ideas is allowed, but the submitted write-up must be individual. Cite any collaborative help.
- **Academic honesty:** Properly cite all sources. Plagiarism will be handled per university policy.

Notes and tips

- Focus on clarity and justification rather than attempting a fully detailed production deployment.
- Explore as more as you want, there is no limitations of what can be considered right or wrong.